BENDAVID MAIMON • 9250 Gilman Dr • La Jolla, CA • bendavid.maimon@gmail.com • (858) 642 5994

Education:

University Of California San Diego **La Jolla, California |*2023-2027*|**

Mechanical engineering - robotics and control {graduating June 2027}

Relevant coursework: Solid mechanics, Fluid Mechanics, Statistical methods, Linear control, Signals, Computational methods, Thermodynamics, Manual and CNC machining, Mechanics of materials. Statics, Dynamics and Vibrations.

Experience:

UCSD Morimoto Robotics Lab - Undergraduate Researcher **La Jolla, California |*2025-current*|**

- **Designed nitinol wristed robotic instruments** for non-invasive surgical removal of **cancerous colonic polyps.**
- **Fabricated** robots using **CNC milling**, then redesigned our production workflow using **sinker-EDM** HSS jigging and **wire-EDM** to reduce machining time **from 4 hours to 23 minutes.**
- Integrated our **control** scheme, implementing **Micro-ROS** to actuate our robot, and programmed a **virtual machine** based **TCP-ROS bridge** to collect high frequency timesynced data between inline load cells, 6 axis load sensors, and EM position data.
- **Designed and fabricated** a sub millimeter Electro surgical knife, with demonstrable use in endoscopic submucosal dissection procedures.
- **Authoring** an academic publication on minimally invasive traction assisted polypectomy, to be published and to be presented at the **IEEE international symposium on medical robotics.**

UCSD Delson Engineering Lab - Undergraduate Researcher **La Jolla, California |*2023-2024*|**

- **Designed** external colonoscope manipulation tools, reducing stress/risk during operation.
- **Researched** advanced tribology techniques to **analyze** friction and perfect the system.
- **Designed** and assembled a portable testing rig for collection of torque and **EMG data** on the advanced colonoscopy simulator in the Center for Surgery of the Future.

Triton Robotics - Mechanical design engineer **La Jolla, California |*2023-2025*|**

- **Redesigned and fabricated** a modular autonomous robotics system for competition in the ARC robotics contest. stripped the previous iteration to **reduce volume by 28%** and increase speed, and **eliminated backlash** by implementing **capstan based actuators.**
- Used **Nvidia jetson** and **Cuda** to fully automate our **control scheme**, enabling fully autonomous targeting, task based repositioning, and real time projectile calculations.
- Contributed to our **3rd place trophy at ARC nationals.**

TOM at UCSD - Founder **La Jolla California |2024-2025|**

- **Founded** the UCSD TOM organization, working with the disabled and elderly in San Diego to **design** custom open source solutions for disabilities that were neglected due to unprofitability.
- Partnered with the Rady children's hospital to design and fabricate **8 3D printed wheelchairs** for children with spina bifida, resulting in 3 additional wheelchair requests from families in the San Diego community
- Collaborated with JSOE to **design** a low cost **open source running blade**, increasing independence and removing the barrier of entry for para-abled sports.

Giesel Library - CDM Operations Specialist **La Jolla, California |*2023-2025*|**

- **Designed** an automated tracking system for patron items, created an **Arduino IoT** library to communicate status and user data with the team via the **slack API.**
- **Programmed** an automated database handling system communicating between Google suite and Ex-Libris.

Skills:

CAD [SolidWorks, fusion 360, Autodesk, OnShape], GD&T, Manufacturing [CNC mill and lathe,sinker-EDM, wire-EDM, water jet, 3D printing, laser cutting, Mill and lathe], programming [java, python, C++, Arduino, MATLAB, ROS],IoT, FEA [Ansys Discovery and Maxwell], control systems, , organizational skills, adaptability, problem solving, project management, Data interpretation, and research skills.